24

Randomised Linear Algebra

Historical Note

William Johnson and Joram Lindenstrauss proved the lemma of Section 24.2 in 1984, as a technical step in a paper about extending Lipschitz maps between Banach spaces. They were not thinking about algorithms and the lemma sat unused by computer science for a decade. What changed was the arrival of matrices too large to look at twice. Frieze, Kannan and Vempala showed in 1998 that a low-rank approximation could be built from a random sample of the rows; Sarlós connected sketching to least squares in 2006; and the 2011 survey of Halko, Martinsson and Tropp assembled the randomised singular value decomposition into the form that is now standard library code. The idea that one should deliberately throw information away, at random, in order to compute faster is younger than most of the theorems in this book, and it is the only genuinely new algorithmic idea in numerical linear algebra since the Krylov methods of the 1950s.

Chapter 17 ended by saying that the escape from the cubic wall is never a better constant, but to stop forming the matrix at all, and it listed sparsity, symmetry, low rank and a fast product. Chapter 18 developed the Krylov methods that exploit them. There is a fourth escape that neither chapter mentioned: replace A by a much smaller random projection of it, solve the small problem, and accept an answer that is nearly right with high probability. This chapter is about when that works and what "nearly" means.

24.1 Projections preserve length on average

Definition 24.1 (Gaussian sketch)**.** *A* Gaussian sketching matrix *is* $S = G/\sqrt{k}$*, where* $G \in \mathbb{R}^{k \times m}$ *has entries drawn independently from the standard normal distribution. Applying* S *to a vector of length* m *produces one of length* k*, and* k *will be very much smaller than* m*.*

Proposition 24.2 (Sketching is an isometry in expectation)**.** *For every fixed $x \in \mathbb{R}^m$,* $\mathsf{E}\left[\|Sx\|^2\right] = \|x\|^2$.

Proof. Write $g_1, \dots, g_k$ for the rows of G, so that $\|Gx\|^2 = \sum_i (g_i^t x)^2$. Each row has $\mathsf{E}[g_i g_i^t] = \mathbf{1}$, since its entries are independent with mean 0 and variance 1, so

$$\mathsf{E}\left[(g_i^t x)^2\right] \;=\; \mathsf{E}\left[x^t g_i g_i^t x\right] \;=\; x^t\, \mathsf{E}[g_i g_i^t]\, x \;=\; \|x\|^2 .$$

Summing over the k rows gives $\mathsf{E}\,\|Gx\|^2 = k\,\|x\|^2$, and dividing by k gives the claim. □

Key Idea

That computation is pure linear algebra and it is the whole reason any of this works: the only property of the distribution used is $\mathsf{E}[gg^t] = \mathbf{1}$, which says the rows are isotropic. What it does *not* say is that $\|Sx\|$ is close to $\|x\|$ for any particular x, only that it is right on average. Everything else in this chapter is the work of upgrading an expectation into a statement that holds with high probability, and then of paying the union bound so that it holds for many vectors at once.

Figure 24.1 says what a sketch is for in one picture. The expensive dimension of a least squares problem is the number of observations, not the number of unknowns, and a sketch removes it: SA keeps all n columns and has only k rows.

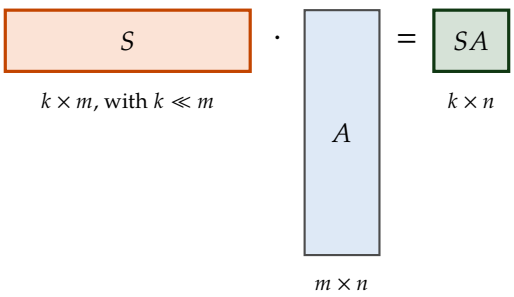


Figure 24.1. What a sketch is for. The expensive dimension of a least squares problem is the number of observations m, not the number of unknowns n, and a sketch removes it: SA has the same n columns and only k rows. The whole question is how small k can be while the answer computed from SA remains usable, and Theorem 24.8 answers it.

24.2 The Johnson and Lindenstrauss lemma

The upgrade from expectation to probability is the one place in this chapter where the input is not linear algebra. We quote it.

Assumption 24.3 (Concentration, quoted)**.** *For a Gaussian sketch S with k rows, every fixed x and every $\varepsilon \in (0,1)$,*

$$\Pr\Big[\,\big|\|Sx\|^2 - \|x\|^2\big| > \varepsilon\,\|x\|^2\,\Big] \;\le\; 2\exp\!\big(-k(\varepsilon^2-\varepsilon^3)/4\big).$$

Theorem 24.4 (Johnson and Lindenstrauss, 1984)**.** *Let $x_1,\dots,x_N \in \mathbb{R}^m$ and $\varepsilon \in (0,1)$. If*

$$k \;\ge\; \frac{16\ln N}{\varepsilon^2-\varepsilon^3}$$

then with probability at least $1-1/N^2$ a Gaussian sketch S satisfies

$$(1-\varepsilon)\,\|x_i-x_j\|^2 \;\le\; \|Sx_i-Sx_j\|^2 \;\le\; (1+\varepsilon)\,\|x_i-x_j\|^2 \qquad \text{for every pair } i,j.$$

Proof. S is linear, so $Sx_i - Sx_j = S(x_i-x_j)$ and the claim is about the $\binom{N}{2}$ difference vectors. Apply Assumption 24.3 to each and take a union bound: the probability that any one of them fails is at most

$$\binom{N}{2}\cdot 2\exp\!\big(-k(\varepsilon^2-\varepsilon^3)/4\big) \;\le\; N^2\exp\!\big(-k(\varepsilon^2-\varepsilon^3)/4\big) \;\le\; N^2\cdot N^{-4} \;=\; N^{-2},$$

using the stated bound on k in the last step. □

Key Idea

Read the bound and notice what is missing from it. The target dimension k depends on the number of points and on the accuracy, and *not at all* on m: a million points in $\mathbb{R}^{1000}$ and a million points in $\mathbb{R}^{10^9}$ need the same k. Distances between finitely many points simply do not require many dimensions to record, and the reason is that N points span a subspace of dimension at most N, on which the geometry lives. The only role of linear algebra in the proof is the first sentence, that S is linear so the pairwise statement reduces to a statement about single vectors; everything else is a union bound.

Caution

The constant is the weak point. At $N = 10^6$ and $\varepsilon = 0.2$ the theorem asks for $k \approx 6910$, which is larger than the ambient dimension of many real problems, so the bound as stated would be useless there. In practice sketches an order of magnitude smaller work well, and the gap between what is provable and what is observed is genuine and not merely a matter of sharpening constants. Treat Theorem 24.4 as an existence statement about the shape of the dependence, which is the part that is robust, rather than as a recipe for choosing k.

24.3 Subspace embeddings

Preserving finitely many distances is not enough to solve a least squares problem, because the candidate solutions form a continuum. What is needed is a sketch that preserves norms on a whole subspace at once.

Definition 24.5 (Subspace embedding). *S is an ε-*subspace embedding *for a subspace $U \subseteq \mathbb{R}^m$ if*

$$(1-\varepsilon)\,\|u\| \;\le\; \|Su\| \;\le\; (1+\varepsilon)\,\|u\| \qquad \text{for every } u \in U.$$

Proposition 24.6 (It is a statement about singular values). *Let the columns of $Q \in \mathbb{R}^{m\times d}$ be an orthonormal basis of U, as supplied by Theorem 12.12. Then S is an ε-subspace embedding for U if and only if every singular value of SQ lies in $[1-\varepsilon,\, 1+\varepsilon]$.*

Proof. Every $u \in U$ is Qy for a unique $y \in \mathbb{R}^d$, and $\|Qy\| = \|y\|$ because Q has orthonormal columns. So the condition reads $(1-\varepsilon)\,\|y\| \le \|SQy\| \le (1+\varepsilon)\,\|y\|$ for all y, and by Proposition 16.9 the extreme values of $\|SQy\|/\|y\|$ are exactly $\sigma_{\max}(SQ)$ and $\sigma_{\min}(SQ)$. □

Figure 24.2 draws the conclusion. Restricted to the subspace, S carries the unit sphere to an ellipsoid whose semi-axes are the singular values of SQ, so being an embedding is exactly the statement that every one of them lies in $[1-\varepsilon, 1+\varepsilon]$.

Assumption 24.7 (Embedding dimension, quoted). *There is an absolute constant c such that a Gaussian sketch with $k \ge c\,(d + \ln(1/\delta))/\varepsilon^2$ rows is an ε-subspace embedding for any fixed d-dimensional subspace with probability at least $1-\delta$.*

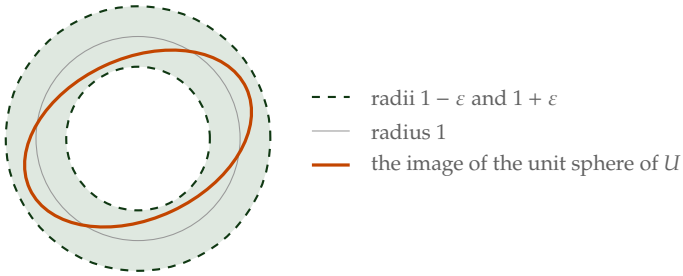


Figure 24.2. Definition 24.5 and Proposition 24.6. Restricted to a d-dimensional subspace, S acts as in Figure 16.2: it carries the unit sphere to an ellipsoid whose semi-axes are the singular values of SQ. Being an embedding means every one of those d semi-axes lies within ε of 1, which is a much stronger demand than preserving a few distances and is why k must now grow with d. Chapter 25 computes how fast.

The dependence $k \sim d/\varepsilon^2$ is not an artefact of the proof, and Chapter 25 will explain exactly where it comes from: it is the width of the Marchenko and Pastur band for a $k \times d$ random matrix. What matters here is that k depends on the dimension of the subspace and not on m.

24.4 Sketch and solve

Theorem 24.8 (Sketched least squares). *Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, let U be the span of the columns of A together with b, and suppose S is an ε-subspace embedding for U with $\varepsilon < 1$. Let $\tilde{x}$ minimise $\|SAx - Sb\|$ and let x^* minimise $\|Ax - b\|$. Then*

$$\|A\tilde{x} - b\| \;\le\; \frac{1+\varepsilon}{1-\varepsilon}\;\|Ax^* - b\|\,.$$

Proof. Both $A\tilde{x} - b$ and $Ax^* - b$ lie in U, so the embedding inequalities apply to them. Using the lower bound, then the fact that $\tilde{x}$ minimises the sketched objective, then the upper bound,

$$\|A\tilde{x} - b\| \;\le\; \frac{\|S(A\tilde{x} - b)\|}{1-\varepsilon} \;\le\; \frac{\|S(Ax^* - b)\|}{1-\varepsilon} \;\le\; \frac{1+\varepsilon}{1-\varepsilon}\,\|Ax^* - b\|\,. \qquad \square$$

Caution

The conclusion is about the *residual*, not about the solution. Nothing in Theorem 24.8 says that $\tilde{x}$ is close to x^*, and when A is ill-conditioned it need not be: by Theorem 17.4 a nearly optimal residual is consistent with a solution that is wrong

in its leading digits. If the coefficients themselves are the object of interest rather than the fit, sketching should be used to build a preconditioner in the sense of Definition 18.9 and the problem then solved iteratively, which gives the accuracy of the exact method at close to the cost of the sketched one.

Caution

A Gaussian sketch is usually the wrong sketch. Forming SA costs kmn multiplications, and solving the original problem by Householder QR costs about $2mn^2$, so a dense Gaussian sketch is slower than doing the work honestly unless $k < 2n$, which the embedding bound forbids. Randomised least squares is only worthwhile with a *structured* sketch that can be applied fast: a subsampled randomised Hadamard transform, which costs $O(mn \log k)$ because the transform is a fast one, or a count sketch, which touches each nonzero entry of A once and so costs $O(\operatorname{nnz}(A))$. The theory above applies to those too, with worse constants. The Gaussian case is the one to reason with and the one not to run.

24.5 The randomised singular value decomposition

The most used of these methods approximates the dominant singular subspace rather than solving a system. It rests on one observation: multiplying A by a random vector produces a random linear combination of its columns, which is very unlikely to be deficient in the directions where A is large.

How many random vectors is a question with a sharp answer, and Figure 24.3 plots it: drawing a few more than the rank wanted buys most of what oversampling can buy. At $k = 10$, five extra samples bring the factor to 1.87 and ten bring it to 1.45, by which point the curve has visibly flattened.

Assumption 24.9 (Randomised range finder, quoted)**.** *Let A have singular values $\sigma_1 \geq \sigma_2 \geq \cdots$, fix a target rank r and an oversampling parameter $p \geq 2$, and let Q be as in Figure 24.4 with $\ell = r + p$. Then*

$$\mathsf{E}\,\left\|A - QQ^{t}A\right\|_{F} \;\leq\; \left(1 + \frac{r}{p-1}\right)^{1/2} \left(\sum_{j>r} \sigma_j^2\right)^{1/2}.$$

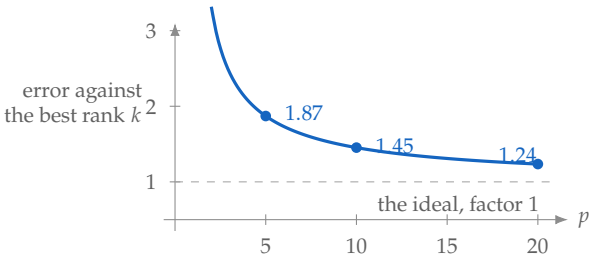


Figure 24.3. Why the randomised range finder oversamples, and why it stops at five. Drawing $k + p$ random test vectors instead of k gives an expected Frobenius error of at most $\big(1 + k/(p-1)\big)^{1/2}$ times the best possible rank-k error, which is the factor of Assumption 24.9. At $k = 10$ that factor is 3.32 with two extra samples, 1.87 with five and 1.45 with ten, and the curve has visibly flattened by then. The cost of each extra sample is one more matrix-vector product, so the practical recommendation of five to ten is not folklore but the point where the curve stops repaying. What no amount of oversampling removes is the gap itself: the factor exceeds one for every finite p, so the method is not a way to compute the exact truncated decomposition of Chapter 16, and was never meant to be.

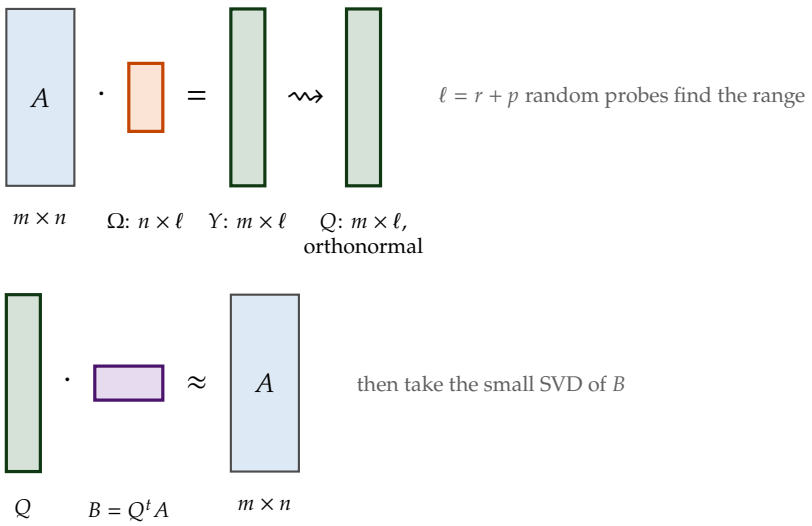


Figure 24.4. The randomised range finder and the decomposition built from it. Only the first row touches the large matrix in a way that depends on the target rank: $Y = A\Omega$ costs $mn\ell$, the orthonormalisation costs $m\ell^2$, and $B = Q^t A$ costs $mn\ell$ again. The total is $O(mn\ell)$ against the $O(mn \min(m, n))$ of a full decomposition, and for ℓ in the tens on a matrix of size 10^5 that is the difference between a second and a week.

Key Idea

The second factor is exactly the error of the best rank-r approximation in the Frobenius norm, by Theorem 16.12. So the randomised answer is optimal up to the first factor, which at $r = 10$ and $p = 5$ is $\sqrt{3.5} \approx 1.87$. Five extra random probes buy a result within a factor of two of the best possible, computed in one pass over the matrix. That is the whole method, and the reason oversampling matters so much for so little cost is that p appears in a denominator.

Example 24.10 (Power iteration when the spectrum is flat)**.** *The bound is weak when the tail $\sigma_{r+1}, \sigma_{r+2}, \ldots$ decays slowly, because then the dominant subspace is not well separated and random probes see little of it. The remedy is to sample from $(AA^t)^q A$ instead of A. By Theorem 16.3 that matrix has the same singular vectors as A and singular values σ_j^{2q+1}, so the ratio σ_{r+1}/σ_r is replaced by its $(2q+1)$st power and the separation improves geometrically. This is the power method of Section 9.6 in a new costume, and $q = 1$ or 2 is usually enough. The price is $2q$ extra passes over A, and the numerical caution of Chapter 17 applies: the intermediate products must be reorthogonalised or the small singular directions are lost to rounding.*

Insight

It is worth naming what has changed. Every algorithm before this chapter was deterministic and produced, up to rounding, the answer. These produce an answer that is right with probability $1 - \delta$, and δ is a parameter one pays for in k. That is a genuine weakening, but two things soften it. The failure probability is exponentially small in the sketch size, so a modest increase buys near-certainty; and the answer is nearly always checkable, since computing $\|A\tilde{x} - b\|$ or $\left\|A - QQ^tA\right\|$ costs one matrix product and settles the question. A randomised method with a cheap certificate is not really a gamble. It is a fast heuristic with a proof attached.

Open Problem 24.11 (The optimal sparse embedding)**.** *A subspace embedding for a d-dimensional subspace of $\mathbb{R}^n$ is a matrix S with m rows preserving all norms on that subspace to within $1 \pm \varepsilon$. Two costs matter: m, which sets the size of the small problem, and the number of nonzeros per column of S, written s, which sets the cost of applying it.* What is the optimal tradeoff between m and s?
What is known. *With $s = m$, that is a dense Gaussian sketch, $m = \Theta(d/\varepsilon^2)$ is necessary and sufficient. At the other extreme, $s = 1$ requires $m = \Omega(d^2)$, so extreme sparsity is*

expensive. Nelson and Nguyên's OSNAP construction achieves $m = O(d\,\mathrm{polylog}(d)/\varepsilon^2)$ with $s = O(\mathrm{polylog}(d)/\varepsilon)$ [480], and subsequent work has improved the logarithms. The exact tradeoff curve is not known, *and in particular whether $m = O(d/\varepsilon^2)$ is achievable with $s = O(1/\varepsilon)$ is open.*

Why the question is delicate. *The failure mode of a sparse sketch is collision: two heavy coordinates of a vector landing in the same row and cancelling. A vector spread evenly across its coordinates is safe, and a spiky vector is not, which is exactly the spikiness of Chapter 47. The proofs therefore split the subspace into a spread part and a spiky part and handle them separately, and the logarithms come from the seam between the two cases. Removing them means finding an argument that does not split.*

Where a reader might start. *Take $d = 1$ and $s = 1$, so S hashes n coordinates into m buckets with random signs. The variance of $\|Sx\|^2$ can be computed exactly:*

$$\mathrm{Var}\big(\|Sx\|^2\big) \;=\; \frac{2}{m}\Big(\|x\|_2^4 - \|x\|_4^4\Big).$$

Read the sign. The fourth norm enters negatively, *so the spiky vector has the* smaller *variance, and the second moment therefore points the wrong way. Take x with two equal nonzero coordinates: with probability $1/m$ they land in the same bucket, and $\|Sx\|^2$ is then 0 or 2 instead of 1, a total failure that no variance can see.* The gap between the variance and the tail is the whole story, *and it is what forces the proofs to split a subspace into a spread part and a spiky part and argue about them separately.*

Example 24.12 (How many rows does the sketch need?)**.** *Let S be $m \times n$ with independent $N(0, 1/m)$ entries. For a fixed x, $\mathsf{E}\,\|Sx\|^2 = \|x\|^2$, and because $m\,\|Sx\|^2/\|x\|^2$ is a chi-squared variable with m degrees of freedom, its variance is $2m$, so*

$$\frac{\mathrm{Var}\big(\|Sx\|^2\big)}{\|x\|^4} \;=\; \frac{2}{m}.$$

At $m = 100$ the standard deviation of $\|Sx\|^2/\|x\|^2$ is $\sqrt{0.02} \approx 0.141$, so a single vector's norm is preserved to about fourteen per cent. That sounds poor, and it is the whole reason the theorem needs a tail bound rather than a variance: to get $\varepsilon = 0.1$ with failure probability 0.01, the single-vector form of the bound asks for $m \geq (8/\varepsilon^2)\log(2/\delta) \approx 4239$ rows, and halving ε to 0.05 quadruples that to about $16\,955$.

Two things are worth reading off. The count does not mention n at all, which is the entire content of the lemma: you may start in a million dimensions and still finish in four thousand. And the dependence is ε^{-2}, not ε^{-1}, so accuracy is expensive in a way

that dimension is not. Sketching is cheap when you can tolerate ten per cent and ruinous when you need one.

Exercises

Exercise 24.13. *Prove Proposition 24.2 again for a sketch whose entries are independent and take the values ± 1 with probability $1/2$ each, scaled by $1/\sqrt{k}$. Which step of the proof changes, and which does not?*

Exercise 24.14. *Evaluate the bound of Theorem 24.4 for $N = 10^4$ and $\varepsilon = 0.1, 0.3, 0.5$. Then hold $\varepsilon = 0.3$ and tabulate k for $N = 10^2, 10^4, 10^6$. Which of the two parameters is expensive, and why does the shape of the dependence explain the name "dimension reduction"?*

Exercise 24.15. *Show that the union bound is doing real work: exhibit a fixed S with $k = 1$ that preserves the norm of one given vector exactly and destroys the norm of another. Deduce that no bound holding for all pairs can avoid a dependence on N.*

Exercise 24.16. *Let Q have orthonormal columns and let S be an ε-subspace embedding for its column space. Show that $\left\| (SQ)^{t}(SQ) - \mathbf{1} \right\|_2 \le 2\varepsilon + \varepsilon^2$, and conversely that this bound with $\varepsilon < 1$ implies the embedding property with a slightly worse constant. Which direction uses Proposition 24.6?*

Exercise 24.17. *Take $A = \begin{pmatrix} 1 & 1 \\ 0 & \eta \\ 0 & 0 \end{pmatrix}$ with η small and $b = (0, 0, 1)$. Compute x^{*}, and find an $\tilde{x}$ whose residual is within 1% of optimal but whose distance from x^{*} is large. Relate the size of the discrepancy to $\kappa(A)$ and to the Warning after Theorem 24.8.*

Exercise 24.18. *For $m = 10^7$ and $n = 500$, count the multiplications in Householder QR on A, in forming SA with a dense Gaussian S of $k = 2000$ rows, and in the QR of SA. Confirm the claim of the second Warning, and say what $\operatorname{nnz}(A)$ would have to be for a count sketch to win.*

Exercise 24.19. *Plot or tabulate the factor* $(1 + r/(p-1))^{1/2}$ *of Assumption 24.9 for* $r = 20$ *and* $p = 2, 5, 10, 20$*. At what point does further oversampling stop paying, and how does the cost of* p *compare with the cost of* r *in Figure 24.4?*

Exercise 24.20. *For* $\sigma_j = j^{-1/2}$ *compute* σ_{11}/σ_{10}*, and then the same ratio for the singular values of* $(AA^t)^q A$ *with* $q = 1$ *and* $q = 2$*. Explain in one sentence why Example 24.10 helps most for exactly this kind of spectrum.*

Exercise 24.21. *Given* Q *from the range finder, show that* $\|A - QQ^tA\|_F^2 = \|A\|_F^2 - \|Q^tA\|_F^2$*, so the error can be certified with one extra product. Why is the analogous check for Theorem 24.8 even cheaper?*

References and further reading

Halko, Martinsson and Tropp [292] is the survey that made these methods standard and is the source of Assumption 24.9 and of the analysis of oversampling; it is unusually well written and should be read before anything else here. Martinsson and Tropp [455] is the modern successor and covers the structured sketches that the second Warning insists on. For sketching applied to regression, Woodruff's monograph [654] is the standard reference and proves Assumption 24.7 along with the count sketch results. Vershynin [627] supplies the probability that this chapter quoted, including Assumption 24.3, and is the right place to go for a reader who wants the concentration arguments rather than their consequences. The original Johnson and Lindenstrauss paper is short and is worth looking at for how little it claims. Trefethen and Bau [607] do not treat randomisation, which dates them, but remain the best account of the deterministic algorithms these methods are competing with.